StockSense Telemetry System

Costed Bill of Materials (BOM) — Hardware Revision v1.1

Date: 06th November 2026
Currency: LKR (Sri Lankan Rupee)
Board Form Factor: 100mm × 100mm

ACTIVE MODULES & ICS	PASSIVES & HARDWARE	EST. TOTAL UNIT COST
LKR 3,920.00	**LKR 531.00**	**LKR 5,251.00**
ESP32-DevKitC, Sensors, Modules	Power, Diodes, Connectors, Switches	Includes 100x100mm 2-Layer PCB

1. CORE CONTROLLER & SENSOR MODULES

REFDES	DESCRIPTION & SPECIFICATIONS	FOOTPRINT / MPN	QTY	UNIT	TOTAL
U1 (or U2)	ESP32-DevKitC (38-pin version) Development Board	ESP32-DevKitC_Module	1	1,950	1,950
J5, J9, J10	HX711 24-Bit ADC Weighing Sensor Modules (Bin 1-3)	Conn_01x04 (2.54mm)	3	250	750
J8	OLED Display / BME280 I2C Interface Header Socket	Conn_01x04_Socket	1	200	200
J11	SHT30 Temperature & Humidity Sensor Breakout Header	Conn_01x04 (2.54mm)	1	350	350
J12	RC522 RFID Reader SPI Interface Header	Conn_01x08 (2.54mm)	1	320	320
J1	Mini360 MP2307 Buck Converter Module (12V to 5V)	Mini360_MP2307_Module	1	350	350

2. POWER MANAGEMENT & PROTECTION DISCRETES

REFDES	DESCRIPTION & SPECIFICATIONS	FOOTPRINT / MPN	QTY	UNIT	TOTAL
Q2	AO3401A P-Channel MOSFET (-30V, -4.2A Reverse Protection)	SOT-23-3	1	35	35
Q3	BC847 NPN General Purpose Transistor (Buzzer Driver)	SOT-23-3	1	15	15
D5	12V 0.5W Zener Diode (Gate Voltage Protection)	SOD-123 / SOT-23	1	15	15
D6	SMAJ15A TVS Diode 15V Unidirectional Clamp	DO-214AC (SMA)	1	45	45
D_BAT1, D_MAINS1	SS34 Schottky Barrier Diodes (40V, 3A, Low Vf OR-ing)	DO-214AB (SMC)	2	30	60
D9	D_Small (Flyback Protection Diode for Buzzer)	SOD-123	1	10	10

3. PASSIVES, INDICATORS & CONNECTORS

REFDES	DESCRIPTION & SPECIFICATIONS	FOOTPRINT / MPN	QTY	UNIT	TOTAL
R1, R8, R10	1kΩ & 10kΩ 1% 0805 SMD Resistors (Buzzer Bias/Pull-down)	R0805	3	5	15
R2, R3, R6	10kΩ 1% 0805 SMD Resistors (EN, BOOT, Protection)	R0805	3	5	15
R_DIV1, R_DIV2	30kΩ / 10kΩ 1% 0805 Resistors (Voltage Divider Sense)	R0805	2	5	10
R4, R5	4.7kΩ 1% 0805 SMD Resistors (I2C Bus Pull-ups)	R0805	2	5	10
D7, D8	3mm / 0805 SMD Status LEDs (Green Power / Red User Debug)	LED_0805	2	12	24
C1,C4,C6,C9..C12	0.1µF 50V Ceramic Decoupling Capacitors X7R 0805	C0805	8	8	64

REFDES	DESCRIPTION & SPECIFICATIONS	FOOTPRINT / MPN	QTY	UNIT	TOTAL
C3	4.7µF 25V Ceramic Filtering Capacitor 0805	C0805	1	12	12
C5	100µF 25V Electrolytic Capacitor (Low ESR Power Buffer)	Radial / SMD 6.3mm	1	35	35
J3	DC Barrel Power Jack (2.1mm × 5.5mm PCB Mount)	BarrelJack_Horizontal	1	85	85
J4	2-Pin Screw Terminal Block (5.08mm Pitch, Battery Input)	TerminalBlock_5.08mm	1	60	60
SW1, SW2	Tactile Push Buttons 6×6mm (EN Reset & BOOT Select)	SW_PUSH_6mm	2	20	40
BZ1	5V Active Piezo Buzzer Module	Buzzer_12mm	1	50	50
PCB Fabrication	Custom 2-Layer FR4 PCB (Est. per-unit cost on batch of 5)	100×100mm FR4 Green HASL	1	800	800

TOTAL ESTIMATED BOM & PCB FABRICATION COST (LKR) **LKR 5,251.00**

Local Sri Lanka Procurement Notes:

- Prices based on prevailing retail rates from local electronics distributors (Tronic.lk, Senith Electronics, Microchip.lk).
- Passive component prices (0805 resistors & capacitors) are estimated based on cut-tape purchases in small prototyping quantities.
- PCB Fabrication per-unit cost assumes a standard 5-board prototype run (100mm × 100mm size limit) via JLCPCB/PCBWay shipped to Sri Lanka.